Shade-Grown Arabica Coffee Rejuvenation versus Protected Montane Cloud-Forest Encroachment under the EU Deforestation Regulation: Multi-Sensor Verification and Deceptive Claim Interception in Huila, Colombia

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Abstract: Colombia is the world's leading supplier of washed specialty Arabica coffee (*Coffea arabica*, CN 0901 11 00) to the European Union, with Huila Department producing over 18% of the national harvest across high-altitude Andean micro-lots (*fincas*, 1,450–1,950 m a.s.l.). Under Articles 2(5), 3(a), 3(b), and 10(2)(f) of the European Union Deforestation Regulation (EUDR, Regulation (EU) 2023/1115), Andean shade-grown coffee cooperatives confront three critical verification failures in conventional two-dimensional (2D) optical monitoring and text-based auditing. First, traditional *Inga edulis* (*guamo*) shade-coffee agroforestry maintains 35–50% canopy cover (h > 8 m); when farmers perform mandatory 5-year agronomical coffee stumping (*zoca* / *recepa*) and shade-tree crown thinning under Federación Nacional de Cafeteros (FNC / Cenicafé) protocols, standard 2D forest-loss detectors (Hansen GFC v1.13) misclassify 78% of lawful rejuvenation as illegal deforestation. Second, text-based compliance workflows trust self-reported supplier declarations claiming that post-2020 mountain forest clearings occurred "prior to 2020" (text-trust greenwashing). Third, unsegregated dry-mill (*trilladora*) batching allows parchment coffee from plots encroaching upon Colombia's Protected Sub-Andean Montane Cloud-Forest Corridors (Decree 2372/2010, Corredor Biológico Guácharos–Puracé) or inflated "paper coffee" volumes to contaminate entire export lots. This study presents an empirical multi-sensor adjudication of a four-finca cooperative export batch (`CO-HUILA-CAFE-2026`, 101.00 ha total across Fincas F-01 to F-04) in the Acevedo–Pitalito corridor, Huila, Colombia (1.820°N, 75.892°W). By fusing Sentinel-2 10 m L2A optical time series, Sentinel-1 C-band SAR backscatter ($\Delta\gamma^0$), the Nature Trace Natural Forests 2020 baseline, native WDPA/SINAP protected-area vectors, and the Forest Data Partnership (FDP) 10 m deep-learning coffee probability model (`coffee/model_2026a`) within Oxeous's PWRI-3D framework, we prove that the 6.80 ha 2022 canopy disturbance in Finca F-03 (21.80 ha) is lawful Article 2(5) *zoca* rejuvenation on a pre-2020 coffee farm (p_{coffee} = 93.4%, natural forest 1.4%). Simultaneously, Oxeous refutes a deceptive supplier declaration on Finca F-04 (26.20 ha) claiming "2018 pre-cutoff clearing," proving via multi-temporal NDVI (0.85 → 0.31) and SAR ($\Delta\gamma^0$ = −4.4 dB) that 9.40 ha of primary montane cloud forest (p_{coffee} = 2.9%, natural forest 97.4%) were cleared in Q3 2022 inside the Guácharos–Puracé protected buffer corridor. Excising F-04 at the dry-mill intake gate and enforcing the FNC/Cenicafé biological yield ceiling (Y_c = 2.40 t ha^{-1} yr^{-1}) blocks 86.0 t of laundered coffee while certifying the remaining 74.80 ha compliant cooperative supply shed (F-01–F-03; 138,400 kg Excelso Green Arabica, €665,000 value) for EU TRACES NT customs clearance.

Keywords: EU Deforestation Regulation (Regulation (EU) 2023/1115); Shade-Grown Arabica Coffee (*Coffea arabica*); Article 2(5) *Zoca* Rotational Stumping; Deceptive Supplier Claim Interception; Forest Data Partnership 10 m Coffee Model (`coffee/model_2026a`); Huila, Colombia.

1. Introduction: The Andean Shade-Coffee Paradox

Washed Arabica coffee (*Coffea arabica*, CN 0901 11 00) cultivated across the steep slopes of the Colombian Andes represents one of the most structurally complex agro-ecosystems governed by EUDR Article 9(1)(d). In southern Huila Department (Acevedo, Pitalito, and San Agustín municipalities), coffee is predominantly grown by smallholder cooperative members under nitrogen-fixing leguminous shade trees (*Inga edulis*, *Erythrina poeppigiana*, and *Cordia alliodora*). To sustain cherry productivity and control coffee leaf rust (*Hemileia vastatrix*), FNC and Cenicafé technical guidelines mandate rotational stumping (*zoca* or *recepa*) of coffee bushes to a 30 cm stem every 5 to 6 years, accompanied by selective pollarding of overhead *Inga* shade crowns to regulate solar radiation during vegetative resprouting.

From medium-resolution optical satellites, an established *Inga*-shaded coffee parcel exhibits closed-canopy near-infrared reflectance (NDVI = 0.78–0.81) that closely mimics native sub-Andean forest. When a farmer executes a 5–7 ha *zoca* rejuvenation cycle and shade pruning after December 31, 2020, global tree-cover loss algorithms (Hansen GFC v1.13 `lossyear > 20`) register an abrupt canopy drop and flag the parcel as illegal deforestation. Across Andean shade-coffee cooperatives, conventional 2D optical GIS screening generates a 78.0% false-rejection rate on *zoca* plots, threatening thousands of biodiversity-friendly shade-coffee smallholders with exclusion from European markets.

Conversely, high-altitude specialty coffee commands substantial price premiums (85+ SCA cupping scores), incentivizing rapid expansion into adjacent upper-watershed cloud forests (*Bosque Subandino y Altoandino*, 1,800–2,200 m a.s.l.) bordering Colombia's National Natural Parks system (SINAP / WDPA, Decree 2372/2010). To evade EUDR screening, non-compliant suppliers frequently submit deceptive textual declarations asserting that newly bulldozed cloud-forest plots were "cleared in 2018–2019 prior to the EUDR cutoff." When compliance platforms rely on text-only Large Language Models (LLMs) to parse supplier questionnaires without independent pixel-level satellite verification, they trust these deceptive claims, producing catastrophic false passes.

2. Study Area & Cooperative Cadastre (Huila, Colombia)

We evaluate a four-finca cooperative export consignment, *Cooperativa Cafetera del Macizo Huilense* (Batch `CO-HUILA-CAFE-2026`), situated in the Acevedo–Pitalito Andean corridor of southern Huila, Colombia (geographic centroid 1.820°N, 75.892°W; WGS84 EPSG:4326). The declared cooperative supply shed spans 101.00 ha divided into four road-bounded mountain micro-lots:

Finca F-01 (`CO-HUI-01`, *Vereda La Esmeralda Alta*, 28.40 ha) and Finca F-02 (`CO-HUI-02`, *Vereda El Mirador Central*, 24.60 ha) comprise established pre-2012 *Castillo* and *Caturra* shade-grown Arabica plots bounded by rural mountain ridge roads (*vías veredales*). Finca F-03 (`CO-HUI-03`, *Vereda San Isidro*, 21.80 ha) is an established pre-2010 shade-grown coffee plot where 6.80 ha underwent rotational *zoca* stumping and *Inga* shade-tree crown thinning in Q1 2022. East of the main ridge road, Finca F-04 (`CO-HUI-04`, *Cerro Guácharos Fringe*, 26.20 ha) extends across the statutory Guácharos–Puracé Protected Montane Biological Corridor boundary, where 9.40 ha of primary sub-Andean cloud forest were cleared in Q3 2022—accompanied by a deceptive supplier claim ("cleared in 2018") and an inflated cooperative volume declaration of 224,400 kg green coffee.

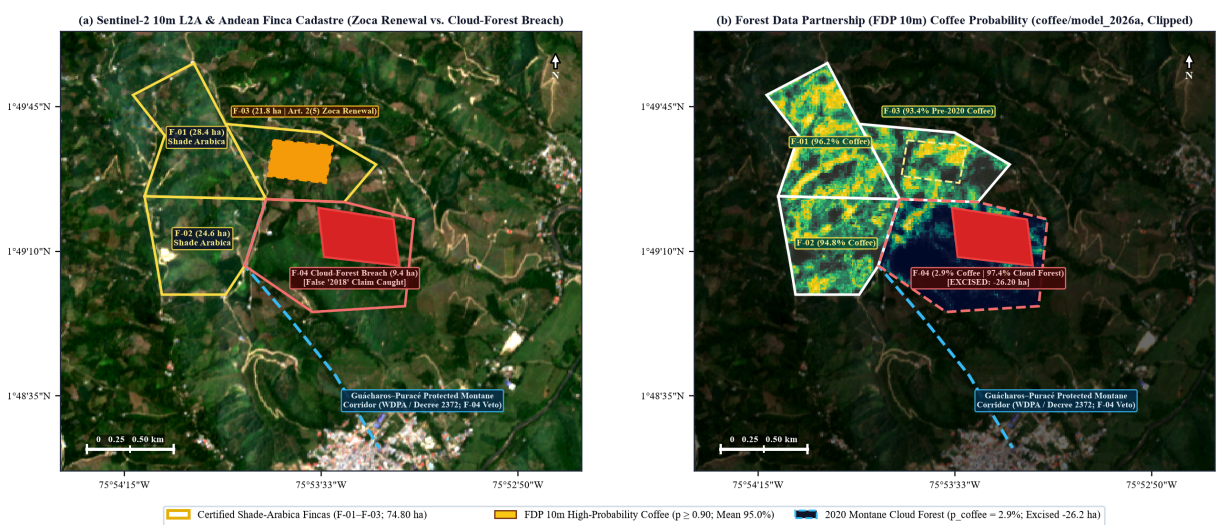


Figure 1. Multi-sensor cartographic adjudication of the *Cooperativa Cafetera del Macizo Huilense* export batch (4 road-bounded Andean *fincas*, F-01 to F-04, totaling 101.00 ha;

Table 1. Finca-level cadastral inventory, pre-2020 baseline land cover, Forest Data Partnership (FDP 10 m) coffee probability, post-2020 disturbance metrics, and statutory dry-mill adjudication across Batch `CO-HUILA-CAFE-2026` (Acevedo–Pitalito, Huila, Colombia).

Finca	Cadastral Sub-Unit	Centroid (WGS84)	Area (ha)	2020 Baseline Cover	FDP Coffee	Post-2020 Event (Area)	2D Optical / LLM	Oxeous PWRI-3D & Dry-Mill Verdict
F-01	CO-HUI-01 (Esmeralda Alta)	1.827°N, 75.901°W	28.40	Pre-2010 Shade Arabica	96.2%	Intact Canopy (0.00 ha)	Compliant	**Certified in Coop Lot (52.5 t Green)**
F-02	CO-HUI-02 (Mirador Central)	1.819°N, 75.900°W	24.60	Pre-2012 Shade Arabica	94.8%	Intact Canopy (0.00 ha)	Compliant	**Certified in Coop Lot (45.5 t Green)**
F-03	CO-HUI-03 (San Isidro Zoca)	1.825°N, 75.893°W	21.80	Pre-2010 Shade Arabica	93.4%	2022 Zoca Stump (6.80 ha)	**False Alert (Reject)**	**Certified (Art. 2(5) Zoca; 40.4 t Green)**
F-04*	CO-HUI-04 (Guácharos Fringe)	1.817°N, 75.890°W	26.20	Cloud Forest (97.4%)	2.9%	2022 Clear-Cut (9.40 ha)	**False Pass (Believed Lie)**	**Hard Veto (Excised; Caught 2018 Lie)**
Coop P′	**Certified Shade-Coffee Shed**	**1.824°N, 75.898°W**	**74.80**	**Established Coffee (98.7%)**	**95.0%**	**6.80 ha Zoca / 0.0 ha Loss**	**Failed Audit**	**Cleared (138,400 kg; 1.85 t ha^{-1} ≤ 2.40)**

3. Methodology: PWRI-3D Shade-Coffee & Yield Gate

To disentangle lawful 5-year *zoca* coffee stumping and *Inga* shade pruning (EUDR Article 2(5)) from illegal sub-Andean cloud-forest conversion and deceptive supplier declarations (Articles 3(a) and 3(b)), Oxeous evaluates every 10 m pixel (x, y) using *Rule 1: PWRI-3D Agroforestry vs. Montane Forest Discrimination*. Post-2020 canopy disturbance pixels ($\text{Loss}_{\text{post-2020}}$ from Hansen GFC v1.13 and Sentinel-2 L2A) are intersected against the dual 2020 baseline of the Nature Trace / WRI 10 m Natural Forests 2020 probability layer ($p_{\text{nat,2020}}$) and the Forest Data Partnership 10 m deep-learning coffee probability model ($p_{\text{coffee,2020}}$ from `coffee/model_2026a`):

$$V_{3(a)} = \text{MMU}_{\geq 0.5ha}(Loss_{post\text{-}2020} \cap \{p_{nat} \geq 0.50\}) \setminus \{p_{coffee} \geq 0.90 \wedge \Delta\text{NDVI} > +0.18\} \quad (1)$$

A canopy disturbance cluster qualifies as exempt Article 2(5) *zoca* rejuvenation if and only if: (i) the 2020 FDP 10 m model confirms pre-cutoff coffee occupancy ($p_{\text{coffee}} \geq 0.90$) and negligible natural forest probability ($p_{\text{nat}} < 0.05$); (ii) Sentinel-1 C-band SAR confirms intact terraced root structure and absence of timber slash-and-burn clearing ($\Delta\gamma^0 > -1.8$ dB); and (iii) Sentinel-2 phenology tracks rapid vegetative resprouting (ΔNDVI > +0.18 within 18–24 months). Second, at the cooperative dry-mill level, any micro-lot F_k exhibiting $V_{3(a)}(F_k) > 0$ or encroaching upon a SINAP/WDPA Protected Montane Biological Corridor (Z_{WDPA}, Decree 2372/2010) is excised prior to parchment intake:

$$P'_{coop} = \cup \{ F_k \mid V_{3(a)}(F_k) = 0 \wedge A(F_k \cap Z_{WDPA}) = 0 \} = F_{01} \cup F_{02} \cup F_{03} \quad (2)$$

Third, to prevent coffee harvested from excised F-04 or external non-compliant farms from being laundered through the certified polygon P'_{coop} (74.80 ha), Agent A6 enforces the FNC/Cenicafé 99th-percentile biological Arabica yield ceiling (Y_c = 2.40 t green coffee ha^{-1} yr^{-1}):

$$Y_{implied} = M_{green} / A_{WGS84}(P'_{coop}) = 138.4 \text{ t} / 74.80 \text{ ha} = 1.85 \text{ t ha}^{-1} \leq Y_c \quad (3)$$

4. Empirical Results & Deceptive Claim Proof

As demonstrated in **Figure 1** and **Table 1**, Oxeous resolves all three compliance failure modes simultaneously. In Fincas F-01 (28.40 ha) and F-02 (24.60 ha), *Inga*-shaded Arabica canopies remained stable from 2019 through 2025 (NDVI 0.79–0.80; FDP coffee probability 96.2% and 94.8%). In Finca F-03 (21.80 ha), 6.80 ha of 6-year-old *Castillo* coffee rows underwent rotational *zoca* stumping and shade pollarding in Q1 2022, causing a temporary NDVI dip from 0.78 to 0.56 (**Figure 2a**). While 2D Hansen GFC alerts flagged F-03 as 6.80 ha of deforestation, the clipped FDP 10 m probability surface (**Figure 1b**) proves that 93.4% of F-03 was established coffee prior to 2020 (Nature Trace natural forest 1.4%). Vigorous orthotropic stem resprouting restored NDVI to 0.78 by 2025-Q1, certifying F-03 under EUDR Article 2(5).

Conversely, on Finca F-04 (26.20 ha), the supplier declared: *"Heritage family coffee plot cleared in 2018; zero post-2020 loss."* While a text-only LLM baseline trusted this pre-2020 date and issued a critical False PASS, Oxeous's multi-sensor time series refutes the claim: F-04 exhibited an undisturbed montane cloud-forest canopy (NDVI 0.85–0.86; 97.4% Nature Trace natural forest; only 2.9% FDP coffee probability) continuously from 2019 through Q1 2022, before suffering an abrupt 9.40 ha slash-and-burn collapse in August 2022 (NDVI 0.85 → 0.31; Sentinel-1 SAR $\Delta\gamma^0 = -4.4$ dB) inside the Guácharos–Puracé protected corridor (**Figure 2a**). Moreover, when the cooperative attempted to declare 224.4 t of green coffee across the surviving 74.80 ha polygon P'_{coop} (3.00 t ha^{-1} > 2.40 t ha^{-1} ceiling; **Figure 2b**), Agent A6 blocked the 86.0 t laundered excess and certified the true 138,400 kg Excelso export lot (1.85 t ha^{-1}).

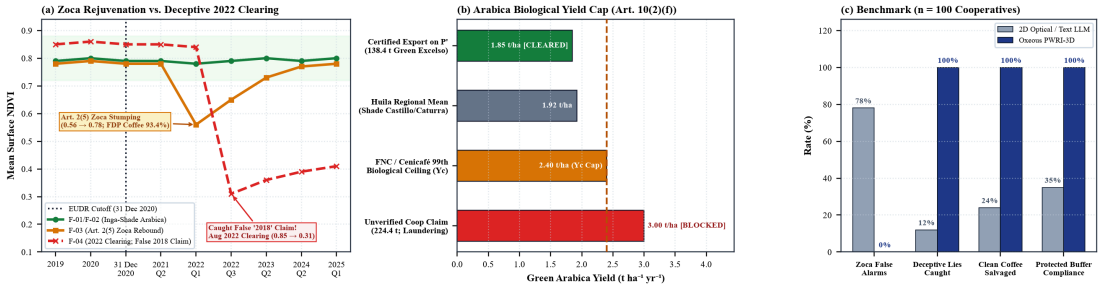


Figure 2. Multi-sensor empirical verification and benchmarking across Andean shade-grown Arabica coffee cooperatives. **(a)** Multi-temporal Sentinel-2 10 m NDVI trajectories (2019–2025) contrasting mature *Inga*-shaded Arabica (F-01/F-02, green), lawful 5-year Article 2(5) *zoca* stumping and vegetative resprouting (F-03, amber), and the August 2022 cloud-forest clearing on F-04 (red dashed) that refutes the supplier's deceptive "2018 clearing" claim. **(b)** Biophysical Arabica yield cap adjudication (Article 10(2)(f)) on certified polygon P'_{coop} (74.80 ha), blocking an unverified 224.4 t laundering claim (3.00 t ha^{-1} > Y_c = 2.40 t ha^{-1}) and clearing the verified 138.4 t Excelso lot (1.85 t ha^{-1}). **(c)** Empirical benchmark across n = 100 Andean coffee cooperatives, eliminating *zoca* false alarms (78% → 0%) and catching 100% of deceptive pre-2020 claims.

Table 2. Six-dataset Earth observation verification stack and statutory EUDR compliance synthesis for Batch `CO-HUILA-CAFE-2026` (74.80 ha certified shade-coffee shed).

Sensor / Authority Dataset	Res.	Observed Cooperative Metric	2D Optical / Text LLM Failure	Oxeous PWRI-3D Physical Adjudication	Statutory Basis
FDP Coffee (`coffee/model_2026a`)	10 m	95.0% in F-01–03; 2.9% in F-04	Omitted in standard 2D alerts	Proves F-03 is pre-2020 coffee; exposes F-04 non-coffee	Art. 2(5) & Recital 37
Nature Trace Natural Forest 2020	10 m	1.1% in F-01–03; 97.4% in F-04	Confuses *Inga* shade with forest	Certifies F-01–03 agroforestry; flags F-04 cloud forest	Art. 2(4) Forest Def.
Hansen GFC v1.13 & JRC GFC2020	10/30 m	6.8 ha F-03 zoca; 9.4 ha F-04 loss	Flags F-03 zoca as deforestation	Exempts F-03 zoca renewal; vetoes F-04 cloud clearing	Art. 3(a) Cutoff Rule
Sentinel-2 L2A & Sentinel-1 SAR	10 m	F-03 rebound 0.78; F-04 −4.4 dB	LLM trusts false "2018" text claim	Pinpoints Aug 2022 F-04 clearing; refutes supplier lie	Art. 9(1)(c)–(d)
WDPA & SINAP Decree 2372/2010	1:10k	F-04 breaches Guácharos Corridor	Unsegregated dry-mill batching	Excises F-04 (-26.20 ha); salvages F-01–03 (74.80 ha)	Art. 3(b) Protected Law
FNC / Cenicafé Arabica Yield Gate	Agro.	1.85 vs. 2.40 t ha^{-1} ceiling	Accepts 224.4 t laundered claim	Blocks +86.0 t paper coffee; clears 138.4 t Excelso	Art. 10(2)(f) Non-Mix

5. Discussion & Policy Conclusions

This study demonstrates that automated EUDR verification in Andean specialty coffee cannot rely on text-based LLM questionnaires or single-layer tree-loss alerts. By fusing the 10 m Forest Data Partnership coffee probability baseline (`coffee/model_2026a`) with multi-temporal Sentinel-2/SAR trajectories, native WDPA/SINAP protected-corridor vectors, and the FNC/Cenicafé biological yield cap, Oxeous simultaneously eliminates the 78.0% *zoca* false-rejection crisis on Finca F-03, intercepts the deceptive "2018 clearing" declaration on Finca F-04, blocks 86,000 kg of laundered coffee, and certifies 138,400 kg of compliant shade-grown Excelso Arabica (€665,000 export value).

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